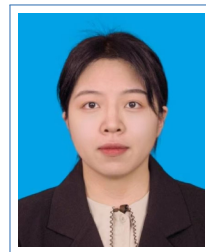


Xiyu Gu

👤 Birthday: 1997.04

☎ +86 186 4359 1667

✉ xiyu.gu@studenti.unipd.it



Education

Jan.2024 – Now

University of Padua

Ph.D. Candidate

Subject: Computing-Communication-Control Codesign

Padua, Italy

Supervisor Prof. Matthias Pezzutto

Co-supervisor Prof. Luca Schenato

June.2022 – Dec.2023.

University of Victoria

Research Assistant

Victoria, Canada

Supervisor Prof. Yang Shi

Sept.2019 – Mar.2022

Northwestern Polytechnical University

Master of Engineering

Subject: Navigation, Guidance & Control

GPA: 3.55/4.00, rank 3/69

Project 985 & 211, Xi'an, China

Supervisor Prof. Jianguo Guo

Co-supervisor Prof. Zongyi Guo

Sept.2015 – Jun.2019

Northwestern Polytechnical University

Bachelor of Engineering

Subject: Flight Vehicle Design & Engineering

GPA: 3.58/4.00, rank 2/32

Project 985 & 211, Xi'an, China

Research Interests

- Control theories, including Model Predictive Control (MPC), prescribed performance control (PPC), robust control, adaptive control, optimal control and cooperative control
- Control over wireless, including edge-based and cloud-based optimal control with packet loss and delay
- Applications, such as multi-performance requirements for flight control, cooperative control of multi-agent system, route planning

Publications

- [1] Y. Dong, Z. Guo, Z. Peng, **X. Gu**, Data-Physics Fusion-Guided Bayesian Fuzzy Learning for Interpretable Reentry Vehicle Trajectory Prediction, *IEEE Transactions on Aerospace and Electronic Systems*, 2026. (Corresponding author).
- [2] **X. Gu**, S. Dey, M. Pezzutto, Cloud-based Reference Governor with Dynamic Computation Offloading over Shared Computing Resources, *European Control Conference (ECC)*, 2026.
- [3] **X. Gu**, L. Schenato, S. Dey, M. Pezzutto, Communication-Computation Trade-off and Optimal Control Selection in LQG Control, *IEEE Control Systems Letters*, 2025.
- [4] **X. Gu**, M. Pezzutto, L. Schenato, S. Dey, Optimal Control Selection over the Edge-Cloud Continuum, *IFAC- PapersOnline*, 2025. (Finalist for Young Author Award)
- [5] **X. Gu**, M. Pezzutto, L. Schenato, S. Dey, Cooperative Edge-Cloud MPC for Constrained Control under Arbitrary Communication Environment, *Automatica (regular paper)*, under review, 2025.

- [6] X. Gu, Z. Guo, X. Wang, et al, Prescribed Performance-Guaranteed Tracking Control: A Switched Positive System Perspective. *IEEE Control Systems Letters*, 2024.
- [7] Z. Guo, X. Gu, Y. Han, et al, Performance-Guaranteed Attitude Tracking Control for RLV: A Finite-Time PPC Approach. *IEEE Transactions on Aerospace and Electronic Systems*, 2024. (**Corresponding author**)
- [8] Z. Guo, D. Herry, X. Gu, et al, Maneuvering tracking algorithm for reentry vehicles with guaranteed prescribed performance, *IEEE Transactions on Aerospace and Electronic Systems*, 2024, 60.3: 3012-3020.
- [9] X. Gu, J. Guo, Z. Guo, Performance improvement-oriented reentry attitude control for reusable launch vehicles with overload constraint, *ISA Transactions*, 2022, 128: 386-396.
- [10] J. Guo, X. Gu, Z. Guo, Asymptotic adaptive tracking control for hypersonic vehicles with guaranteeing multi-performance requirements. *Aerospace Science and Technology*, 2020, 105: 106025.

Awards & Competitions

National Awards of China.....

Postgraduate Admission with exemption of examination	Sept. 2018
The National Encouragement Scholarship	Oct. 2017
The National Encouragement Scholarship	Oct. 2016

School-level Awards.....

The excellent Master's thesis of Northwestern Polytechnical University	Sept. 2022
The Second-class Postgraduate Scholarship	Oct. 2020
The Special scholarship of Wu Yajun	Oct. 2020
The First-class Postgraduate Scholarship	Oct. 2019
The Second-class Undergraduate Scholarship of NWPU	Oct. 2018
The First-class Undergraduate Scholarship of NWPU	Oct. 2017
The First-class Undergraduate Scholarship of NWPU	Oct. 2016

Competitions.....

The Second Prize in Mathematical Contest of Modelling of Shaan'Xi Province	Oct. 2018
The First Prize in Mathematical Contest of Modelling of NWPU	Oct. 2017
The Second Prize of the 11th Higher Mathematics Competition in Shaan'Xi Province	Aug. 2017